

数值分析第十八次作业

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1. 设下列预估-校正格式 (Heun 方法)

$$\begin{cases} \bar{y}_{n+1} = y_n + hf(t_n, y_n), \\ y_{n+1} = y_n + \frac{h}{2}(f(t_n, y_n) + f(t_{n+1}, \bar{y}_{n+1})), \end{cases}$$

的局部截断误差为 $\tau(h) = \mathcal{O}(h^p)$, 求 p .

解答: 简记 $f(t_n, y(t_n)) = f, f_t(t_n, y(t_n)) = f_t$, 其余类似. 由 Taylor 展开可得局部截断误差

$$\begin{aligned} \tau(h) &= y(t_{n+1}) - y(t_n) - \frac{h}{2}(f(t_n, y(t_n)) + f(t_{n+1}, y(t_n) + hf(t_n, y(t_n)))) \\ &= hf + \frac{h^2}{2}(f_t + f_y f) + \frac{h^3}{6}(f_{tt} + 2f_{ty}f + f_{yy}f^2 + f_t f_y + f_y^2 f) \\ &\quad - \frac{h}{2}f - \frac{h}{2}\left(f + hf_t + hf_y f + \frac{1}{2}h^2 f_{tt} + h^2 f_{ty}f + \frac{1}{2}h^2 f_{yy}f^2\right) + \mathcal{O}(h^4) \\ &= \mathcal{O}(h^3), \end{aligned}$$

故 $p = 3$.

□

2. 求证: 梯形公式

$$y_{n+1} = y_n + \frac{h}{2}(f(t_n, y_n) + f(t_{n+1}, y_{n+1}))$$

的局部截断误差为 $\tau(h) = \mathcal{O}(h^3)$.

解答: 由 Taylor 展开可得局部截断误差

$$\begin{aligned} \tau(h) &= y(t_{n+1}) - y(t_n) - \frac{h}{2}(f(t_n, y(t_n)) + f(t_{n+1}, y(t_{n+1}))) \\ &= y(t_{n+1}) - y(t_n) - \frac{h}{2}(y'(t_n) + y'(t_{n+1})) \\ &= hy'(t_n) + \frac{h^2}{2}y''(t_n) + \frac{h^3}{6}y'''(t_n) - \frac{h}{2}\left(y'(t_n) + y'(t_n) + hy''(t_n) + \frac{h^2}{2}y'''(t_n)\right) + \mathcal{O}(h^4) \\ &= \mathcal{O}(h^3). \end{aligned}$$

□

3. 求下列隐式单步法的阶

$$y_{n+1} = y_n + \frac{h}{6} \left(4f(t_n, y_n) + 2f(t_{n+1}, y_{n+1}) + h \frac{d}{dt} f(t, y(t)) \Big|_{t=t_n} \right).$$

解答: 由 Taylor 展开可得局部截断误差

$$\begin{aligned} \tau(h) &= y(t_{n+1}) - y(t_n) - \frac{h}{6} \left(4f(t_n, y(t_n)) + 2f(t_{n+1}, y(t_{n+1})) + h \frac{d}{dt} f(t, y(t)) \Big|_{t=t_n} \right) \\ &= y(t_{n+1}) - y(t_n) - \frac{h}{6} (4y'(t_n) + 2y'(t_{n+1}) + hy''(t_n)) \\ &= hy'(t_n) + \frac{h^2}{2} y''(t_n) + \frac{h^3}{6} y'''(t_n) + \frac{h^4}{24} y''''(t_n) \\ &\quad - \frac{h}{6} \left(4y'(t_n) + 2 \left(y'(t_n) + hy''(t_n) + \frac{h^2}{2} y'''(t_n) + \frac{h^3}{6} y''''(t_n) \right) + hy''(t_n) \right) + \mathcal{O}(h^5) \\ &= \mathcal{O}(h^4), \end{aligned}$$

故阶为 3.

□

4. 考察非等距的时间步长 $h_n = t_{n+1} - t_n > 0, t_n \in [0, T]$. 假设迭代误差满足递推关系

$$|e_{n+1}| \leq (1 + Lh_n)|e_n| + Ch_n^{p+1}, \quad \forall 0 \leq n \leq N-1,$$

求证: 在 $|e_0| \sim \mathcal{O}(h^p)$ 时, 仍然有离散 Gronwall 型不等式误差估计 $|e_n| \sim \mathcal{O}(h^p)$, 其中 $h = \max_{0 \leq k \leq N-1} h_k$ 为分划的尺度.

解答: 注意到 $\forall 0 \leq n \leq N-1$, 都有

$$\begin{aligned} \frac{|e_{n+1}|}{h^p} + \frac{C}{L} &\leq \frac{(1 + Lh_n)|e_n| + Ch_n^{p+1}}{h^p} + \frac{C}{L} \\ &= (1 + Lh_n) \left(\frac{|e_n|}{h^p} + \frac{C}{L} \right) - Ch_n + \frac{Ch_n^{p+1}}{h^p} \\ &\leq (1 + Lh_n) \left(\frac{|e_n|}{h^p} + \frac{C}{L} \right), \end{aligned}$$

故 $\forall 0 \leq n \leq N$, 都有

$$\frac{|e_n|}{h^p} + \frac{C}{L} \leq \prod_{k=0}^{n-1} (1 + Lh_k) \cdot \left(\frac{|e_0|}{h^p} + \frac{C}{L} \right),$$

从而

$$\begin{aligned}|e_n| &\leq h^p \left(\prod_{k=0}^{n-1} (1 + Lh_k) \cdot \left(\frac{|e_0|}{h^p} + \frac{C}{L} \right) - \frac{C}{L} \right) \\&\leq h^p \left(\prod_{k=0}^{n-1} e^{Lh_k} \cdot \left(\frac{|e_0|}{h^p} + \frac{C}{L} \right) - \frac{C}{L} \right) \\&= h^p \left(e^{Lt_n} \left(\frac{|e_0|}{h^p} + \frac{C}{L} \right) - \frac{C}{L} \right) \\&\leq h^p \left(e^{LT} \left(\frac{|e_0|}{h^p} + \frac{C}{L} \right) - \frac{C}{L} \right) \sim \mathcal{O}(h^p).\end{aligned}$$

□